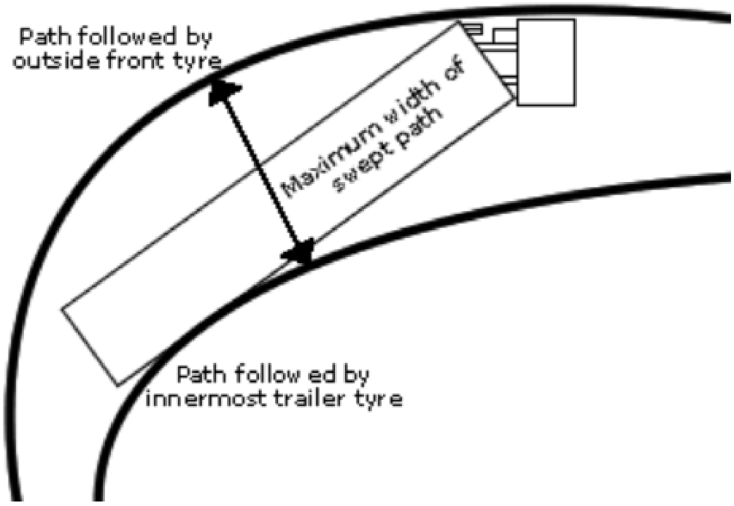



**Figure 2.5:** Swept Path Width measurement for a given vehicle

### 2.5.2 Low Speed Off-tracking

When an articulated commercial vehicle makes a turn at a low speed, the wheels of the rearmost trailer axle follows the path which is inside of the path of the tractor steering axle. This is called low speed off-tracking, as illustrated in Figure 2.6. It indicates the measure of swept path of the vehicle and its lateral road space requirement when the vehicle turns at 90 degree arc. The low speed off-tracking is measured as the maximum radial distance between the path of the midpoint of the steer axle and the path of the midpoint of the rearmost trailer axle.

### 2.5.3 High Speed Off-tracking

When an articulated commercial vehicle makes a steady turn at high speed, the rear of the truck tends to swing outward due to increased lateral acceleration. As the vehicle's speed increases, the low-speed off-tracking diminishes until the rearmost trailer axle follows the same path as the midpoint of the steer axle. At even higher speeds, the rearmost trailer axle begins to follow a path outside the steer axle's midpoint path. This phenomenon is known as high-speed off-tracking, and it is measured by the radial distance between the path of the midpoint of the steer axle and the path of the midpoint of the rearmost trailer axle, as depicted in Figure 2.7.

### 2.5.4 Rearward Amplification

When an LCV makes a sudden lateral movement, each unit in the combination experiences varying levels of lateral acceleration. This lateral acceleration increases progressively towards the rearmost unit of the vehicle, a phenomenon known as rearward amplification [7], as shown in Figure 2.8. Lower values of rearward amplification indicate better performance of the LCV, with an ideal rearward amplification

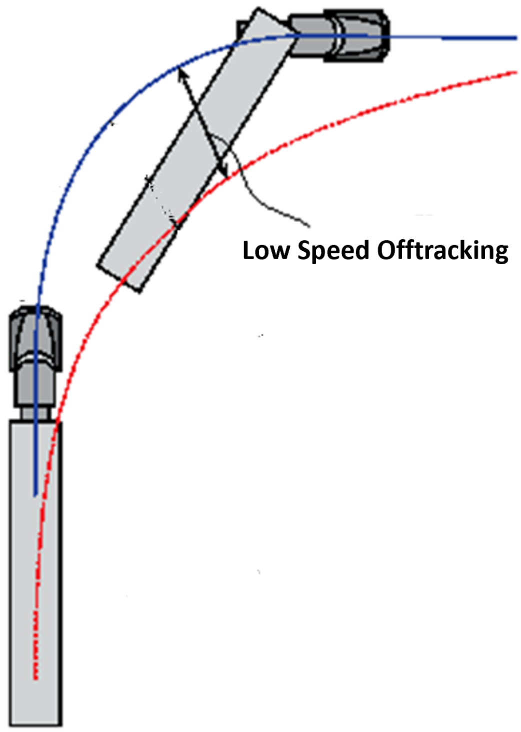



**Figure 2.6:** Lateral off-tracking at low speed

value being one, meaning there is no amplification of lateral acceleration from the front to the rear of the vehicle.

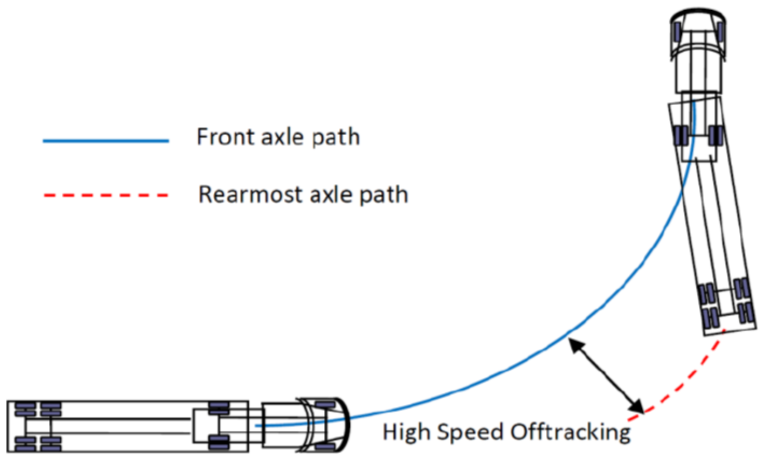



**Figure 2.7:** Lateral off-tracking distance at high speed

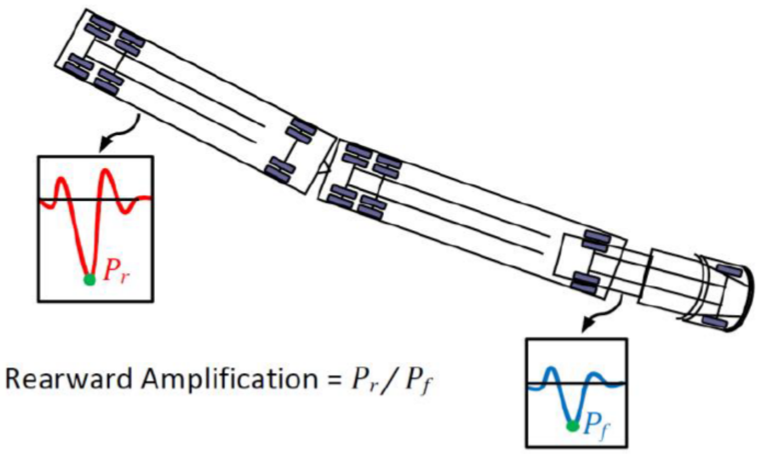



**Figure 2.8:** Rearward amplification of vehicle in terms of yaw rate ($P_f$ and $P_r$)

# 3

# Optimization

This chapter introduces the method for optimization, which can be loosely used for manual drive and automated drive. The overall optimization process is explained along with desired control allocations i.e propulsion and steering is optimized.

## 3.1 Optimal Control

Before getting into the process of optimization, it is crucial to understand why optimized control actions are necessary. Optimal control is the process of determining a control for a dynamic system such that a certain optimality criterion is achieved. This involves finding the best possible way to control a system to achieve desired outcomes while minimizing or maximizing specific performance measures [8]. This could lead to a system that is highly performance oriented and efficient while considering the factor of stability and safety.

## 3.2 Setting up Optimisation

The optimization scheme is broken down into several steps, which was carried out in symbolic tool for numerical optimization called CasADi. This is used for algorithmic differentiation and gradient based numerical optimization with strong focus on optimal control. One of the main advantages that it has is the use of the syntax of Computer Algebra Systems (CAS), allowing the user to construct symbolic expressions that can be differentiated in an efficient way using different algorithms for algorithmic differentiation.

### 3.2.1 System Dynamics

In order to define a system dynamics fixed set of inputs and state vector was chosen. This completely defines the system in explicit way, although being highly non-linear. Hence, the state space model for A-double is described below:

#### 3.2.1.1 Kinematic System Dynamics

To define the system dynamics we need state variable and control actions. We assume that the tractor's CoG location is available from the GPS sensor data. Consequently, we can interpolate the positions of the other units behind the tractor, using their respective yaw angles too, which will be used in our later optimization process. As a result state vector and control vector is defined as follows:

$$X = [X_1, Y_1, \phi_1, \phi_2, \phi_3, \phi_4] \tag{3.1}$$
$$U = [\delta_1, \delta_2] \tag{3.2}$$

where $(X_1, Y_1)$ is the global coordinate of CG location of tractor and $\phi_i$ is the heading angle of respective units.

such that

$$\dot{X} = f(X, U) \tag{3.3}$$

has following representation in explicit form:

$$\dot{X}_1 = v_{1x}\ cos(\phi_1) \tag{3.4}$$

$$\dot{Y}_1 = v_{1x}\ sin(\phi_1) \tag{3.5}$$
$$\dot{\phi}_1 = v_{1x}/R_{11} \tag{3.6}$$
$$\dot{\phi}_2 = v_{2x}/R_{21} \tag{3.7}$$
$$\dot{\phi}_3 = v_{3x}/R_{31} \tag{3.8}$$
$$\dot{\phi}_4 = v_{4x}/R_{41} \tag{3.9}$$

Here, $\dot{\phi}_i$ is nothing but yaw rate of $i_{th}$ unit.

#### 3.2.1.2 Kinetic System Dynamics

Similar, to kinematic system dynamics, state variables of the model are

$$X = [X_1, Y_1, \phi_1, \phi_2, \phi_3, \phi_4, v_{1x}, v_{1y}, \dot{\phi}_1, \dot{\phi}_2, \dot{\phi}_3, \dot{\phi}_4] \tag{3.10}$$

While the control actions are

$$U = [F_{x12v}, \delta_1, \delta_2] \tag{3.11}$$

where, $F_{x12v}$ is the traction force from rear axle of tractor.

#### 3.2.1.3 Time to Path Distance Conversion

At low-speeds, a steering delay is required for dolly steering angles, otherwise tail swing increases. The steering delay for dolly rear axle is derived using geometric relation between axle position.
The system dynamics defined above is in time domain. Due to this the vehicle speeds are not allowed to vary during the turn especially stopping of the vehicle. However, in practical driving situations, e.g. sharp turns in parking lots etc., the driver often change the speeds and even could decide to stop the vehicle completely. Therefore it is important to formulate the problem in path distance domain. According to

[9] conversion from the time domain to the space domain can be performed by performing the following change of variables that results.

$$\frac{d(.)}{dt} = \frac{d(.)}{ds}\frac{ds}{dt} = v_{1x}\frac{d(.)}{ds} \tag{3.12}$$

and

$$\frac{d^2(.)}{dt^2} = \frac{d(.)}{dt}(\frac{d(.)}{ds}\frac{ds}{dt}) = v_{1x}^2\frac{d^2(.)}{ds^2} + v_{1x}\frac{dv_{1x}}{ds}\frac{d(.)}{ds} \tag{3.13}$$

With this knowledge, the system dynamics for both the kinematic and kinetic models (as described in equations 3.2-3.8) were modified and subsequently utilized in the optimization process.

### 3.2.2 Initialization

To solve the nonlinear state space model integration is performed in accordance to the guess values. This also affects the optimization performance and efficiency. Since the optimisation is based on a receding horizon approach and we know the trajectory of the curve or maneuver. This helps in generating fairly accurate initial guess values for the optimal position and yaw angles of the vehicle. The state trajectory is discretized in to N intervals and system dynamics from equations are simulated using a Runge-Kutta integrator of order 4 (RK4) with a number of integration steps defined by 0.5m.

### 3.2.3 Constraints

Good constraints are essential to ensure the optimisation of the desired cost function is carried out within the realistic limitations of the actuators and also ensure the safe operation of the vehicle, not leading to unsafe modes such as jack-knifing or trailer swing. In order to ensure this, the major constraints limits on the steering wheel angle, articulation angle limits and also limit on the maximum force input based on the limit of the traction ellipse for every axle.

#### 3.2.3.1 Kinematic Model Constraints:

Since the kinematic model doesn't involve any forces, the only actuation we have is with respect to the steering angles of the dolly and tractor. The constraints here are with respect to the initial and final positions and angles of the tractor, to ensure the optimisation begins from a desired state. The steering angles of the tractor and dolly axles are also limited to a range of steering wheel angle that is achievable in real life.

#### 3.2.3.2 Kinetic Model Constraints:

With the inclusion of wheel forces in the kinetic model, it becomes to critical to ensure the optimisation follows the traction limits of the tire and hence a constraint is added such that the ability of generating longitudinal and lateral forces at the contact patches for every axle respects the traction ellipse.

### 3.2.4 Cost Function

Using all the state variables and inputs defined above and knowing the co-ordinates of our reference path we are able to define a mathematical expression, which when minimised at every instance in a maneuver leads to the minimisation of our desired characteristic, which in our case is:

- Objective 1 : Swept path minimization at low speed.
- Objective 2 : Minimization of high speed off-tracking.
- Objective 3 : Minimization of rearward amplification.

The expression used for achieving different objectives is the cost function, the cost function can be formulated in multiple ways. Here we explore four different ways of formulating the cost function and then compare their performances. In the cost function, we have two parts one part of the cost function gives us the cost of state deviation and the second part gives us the cost of the controlled input.

The cost of deviation follows an infinity norm constraint that is a variable is assigned with an inequality constraint on the state deviation, which limit the maximum absolute value of state deviation, this basically helps optimization to ensure that no single variable dominates the solution.

#### 3.2.4.1 Cost Function 1: Tractor following approach

This cost function was formulated to replicate a manual driving scenario, where the tractor is controlled by a steering input from the driver. In this scenario the midpoint of the axle (when kinematic model is used) or CoG of the tractor (when kinetic model is used) is used as the reference trajectory for the following units. The cost function is then designed to penalise the deviation of the following units from the reference trajectory i.e. the path traced by the axle midpoint or CoG of tractor.

If the path traced by the axle mid point or CoG of the tractor is given by $(X_{tractor}, Y_{tractor})$ and the position of the following unit's CoG is given by $(X_i, Y_i)$. The deviation $e$ of the following unit from the reference path is calculated accounting for the delay in travelled distance (meaning when both tractor and trailing unit would have been on same location):

$$e_i = ((X_{tractor} - X_i)\ sin(\phi_i) + (Y_{tractor} - Y_i)\ cos(\phi_i))^2 \tag{3.14}$$

such that $e_i < r_{ot}$, where, $r_{ot}$ is the infinity norm variable and $i = 1, 2, 3$ refers to the first trailer, the dolly, and the second trailer, respectively. The cost function is then given by:

$$F = w_1\ \sum(r_{ot}) + w_2\ \sum(U) \tag{3.15}$$

where, $w_1, w_2$ are weight penalization and,
$U = [\delta_1, \delta_2, F_{x12v}]^T$or $[\delta_1, \delta_2]^T$, depending on the vehicle model used.

#### 3.2.4.2 Cost Function 2: Minimizing the deviation of Axle mid-point.

This cost function is aimed more towards autonomous or semi-autonomous driving application, where the control allocator also assigns values to the tractor steering. In this case, the system has trajectory information and from this reference coordinates are generated. The cost function penalises the deviation of axle mid point from this reference path.
In this scenario, the reference coordinates of the path is given by $(X_{path}, Y_{path})$ and the position of the axle mid points is given by $(X_i, Y_i)$. The deviation $e$ of the following unit from the reference path is calculated accounting for the delay in travelled distance:

$$\begin{aligned} e_i = ((X_{path} - X_i)\ sin(\phi_i) + (Y_{path} - Y_i)\ cos(\phi_i))^2 \\ \text{such that} \quad e_i < r_{ot} \end{aligned} \tag{3.16}$$

where, $r_{ot}$ is infinity norm variable and $i$ = 1,2,3,4 refers to the tractor, the first trailer, the dolly and the second trailer, respectively. The cost function is then given by:

$$F = w_1\ \sum(r_{ot}) + w_2\ \sum(U) \tag{3.17}$$

where, again $w_1, w_2$ are weight penalization and,
$U = [\delta_1, \delta_2, F_{x12v}]^T$or $[\delta_1, \delta_2]^T$, depending on the vehicle model used.

#### 3.2.4.3 Cost Function 3: Body corner deviations

This cost function tries to implement the definition of a swept path i.e. the distance between the inner and outer extremities of the vehicle while travelling through a corner. In multi-unit vehicle with multiple actuators, there is a possibility that any of the leading or trailing corner of every unit could form the outer most boundaries of the swept path. Hence, an array of distances is calculated from the the leading edge of the tractor's inner and outer corners to the leading and trailing edges of the following units as shown in the image below.

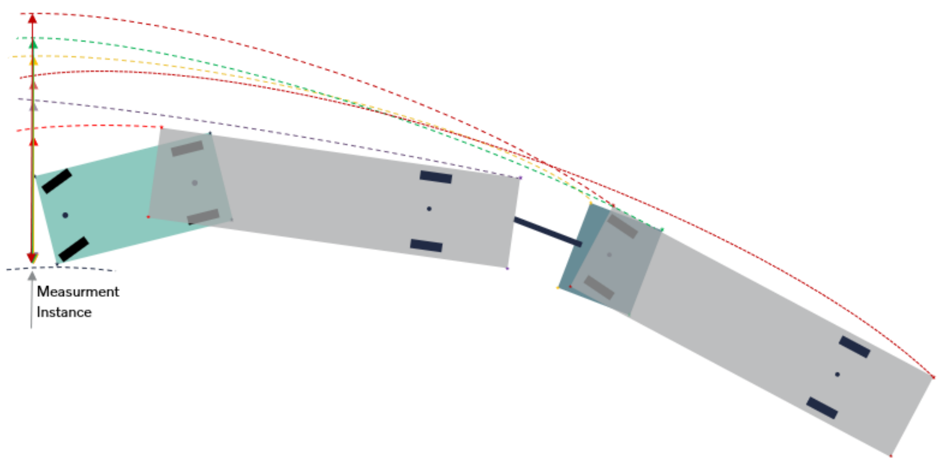



**Figure 3.1:** Calculation of swept path in body point cost function.

If $(X_{tractor_{in}}, Y_{tractorin})$ and $(X_{tractor_{out}}, Y_{tractorout})$ are the leading edges of the tractor body on the inner and outer sides - say left and right. And the inner and outer leading and trailing edges of the following units is given by $(X_{i_{in}}, Y_{iin})$ and $(X_{i_{out}}, Y_{iout})$ respectively.

The swept path measurement from the inner side of the tractor is given by:

$$\begin{aligned} SP_{in} = ((X_{tractor_{in}} - X_{i_{out}})\ sin(\phi_i) + (Y_{tractor_{in}} - Y_{i_{out}})\ cos(\phi_i))^2 \\ \text{such that} \quad SP_{in} < r_{ot1} \end{aligned} \tag{3.18}$$

where, $r_{ot1}$ is infinity norm variable
and $i = 1, 2, 3$ refers to the first trailer, the dolly, and the second trailer, respectively. The swept path measurement from the outer side of the tractor is given by:

$$\begin{aligned} SP_{out} = ((X_{tractor_{out}} - X_{i_{in}})\ sin(\phi_i) + (Y_{tractor_{out}} - Y_{i_{in}})\ cos(\phi_i))^2 \\ \text{such that} \quad SP_{out} < r_{ot2} \end{aligned} \tag{3.19}$$

where, $r_{ot2}$ is another infinity norm variable
Finally, the cost function is then given by:

$$F = w_1 \sum(r_{ot1}) + w_2 \sum(r_{ot2}) + w_3 \sum(U) \tag{3.20}$$

where, $w_1, w_2, w_3$ are weight penalization and, $U = [\delta_1, \delta_2, F_{x12v}]^T$or $[\delta_1, \delta_2]^T$, depending on the vehicle model used.

#### 3.2.4.4 Cost Function 4: Yaw Amplification

This cost function is build to achieve the objective of minimising rearward amplification. As the definition of rearward amplification suggests its basically the ratio of maximum yaw rate of rearmost unit to maximum yaw rate of tractor. But here instead of ratio we take subtraction in order to avoid the possibility of an in-feasibility during optimization.

$$F = w_1 \sum(e) + w_2 \sum(U) \tag{3.21}$$

where, $e_j = \dot{\phi}_{4j} - \dot{\phi}_{1j}$ for $j_{th}$ instance and,
$U = [\delta_1, \delta_2, F_{x12v}]^T$

### 3.2.5 Solution finding or Solver formulation

An NLP solver for CasADi is a vector that takes inputs and state vector with initial guesses for the solution and returns the optimal solution. The most popular NLP solver used in CasADi is IPOPT. Interior Point OPTimizer is a nonlinear optimization algorithm designed to deal with large-scale nonlinear programs. It is an open-source optimization algorithm that uses a interior-point line search filter method. IPOPT allows both objective function and constraints to be nonlinear and non convex as long as all functions are twice continuously differentiable. IPOPT is more accurate then sequential programs since it directly evaluates the nonlinear functions rather then linear or quadratic approximations [10].

## 3.3 Performance Assessment Maneuvers

To evaluate the performance of cost functions in different maneuvering scenarios, the most common road curvatures which challenge maneuverability in long combination vehicle were chosen as follows based on PBS [11].

- 180 degree roundabout maneuver :
  In order to evaluate the swept path and off-tracking at relatively high speed, this particular maneuver was chosen, which also represents the good characteristic of steady state. The radius of curve is set to be 30m with friction of 0.8 and vehicle is made to run at 10 km/hr and 40 km/hr to assess swept path and off tracking respectively.
- 90 degree turning maneuver:
  From Performance Based Characteristic test, the maneuver was obtained to access the A double combination for better performance in terms of swept path at tight corner negotiation at low speed of 10 km/hr and high speed off tracking for radius of 100 m at vehicle speed 75 km/hr.
- S-Curve turning maneuver:
  This curve specifically tests the swept path and steering effort from the driver, to compensate for very short steering input interval change with respect to tighter turn in opposite direction. This curve is evaluated for low speeds of 10km/hr and turning radius in both directions are 20 m.
- Single and double lane change:
  The lane change maneuvers are used to assess the rearward amplification of the LCV while travelling at high speed of 80 km/hr. These kind of maneuvers are usually seen at high speeds during overtaking and obstacle avoidance.

# 4

# Simulation Results

Using above described cost functions, the lateral performance of A-double combination was assessed for given maneuvers. The optimization was done using both the models that is kinematic and Single track model, which also gives a scope to compare between two. Originally, the results depicts the improvement on performance in terms of swept path, off tracking and rearward amplification using active steerable dolly. Further the cost of computation and the robustness of model is evaluated in very crude manner.

## 4.1 Objective 1 Reflection - Minimizing Swept Path Width

### 4.1.1 Kinematic Results

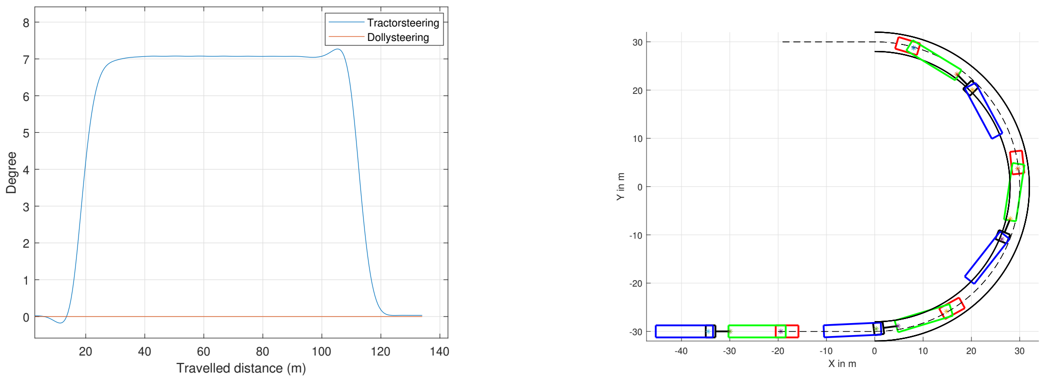
In order to appreciate the benefit of an optimally steered dolly in reducing the swept path width for a given maneuver, it is essential to know the swept path for the combination in the same maneuver without a steerable dolly. This case of vehicle combination with no active steerable dolly is used a baseline. In this case only the tractor is made to follow a certain reference path, without any actuation in the following units. These results are then used to compare the swept path width results obtained by allocating optimal steering angle for the dolly based on the evaluation of different cost functions stated above.

#### 4.1.1.1 180-Degree Turn

The plots in Figure 4.1 show the steering angles for the tractor and dolly and the trajectory of the vehicle combination. This is the base line case, where we have no dolly steering and as a result it can be seen the last unit of the combination has a very high lateral deviation from the path traced by the tractor resulting in a higher swept path. Figures 4.2 to 4.4 depict the steering angles for the tractor and dolly along with the resulting trajectories of the combination for different cost function evaluations.

#### 4.1.1.2 No Dolly Steering

For this baseline case, the combination goes about the 180 degree curve which has a radius of 30m as mentioned earlier, at a constant low speed of 10 m/s. For this maneuver without dolly steering, the combination's maximum swept path width came up to 5.3m.

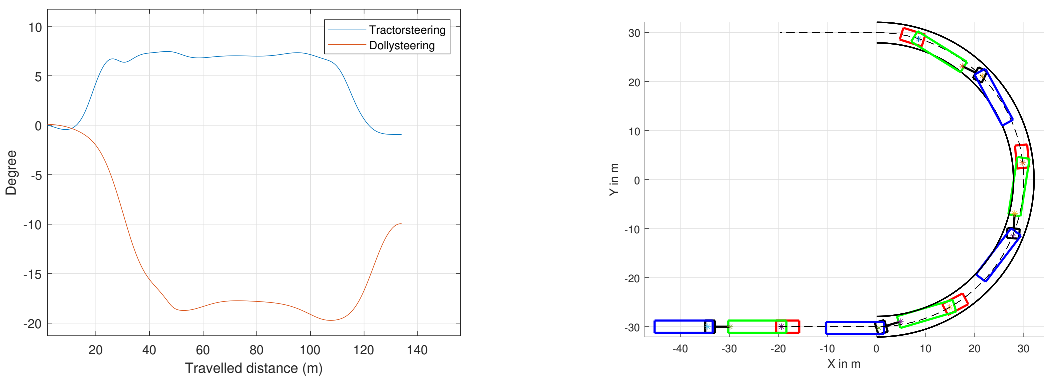



**(a)** Steering control input  **(b)** Vehicle trajectory

**Figure 4.1:** Control inputs and trajectory of vehcile without optimization

#### 4.1.1.3 Dolly Steering - Tractor Following Cost Function

By comparing Figure 4.2 with Figure 4.1, it can be seen the lateral deviation of the last trailer has reduced with the allocation of an optimal steering angle for the dolly. The characteristics of the maneuver and vehicle speed remain the same and the maximum swept path width was reduced to 3.85m



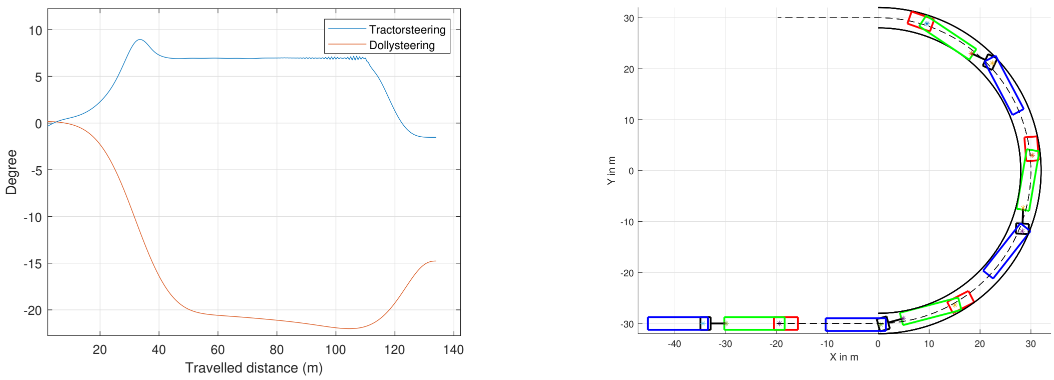

**(a)** Steering control input  **(b)** Vehicle trajectory

**Figure 4.2:** Optimal control inputs and trajectory of vehicle using cost function 1

#### 4.1.1.4 Dolly Steering - Axle Mid-Point Cost Function

In figure 4.3, we observe the performance of the axle mid-point cost function in reducing the swept path width during a 180-degree curve maneuver. In both figures 4.1b and 4.3b, the LVC undergoes the exact same maneuver. It is evident that the cost function effectively reduces the deviation of the trailing units in the latter case, keeping them well within the road boundaries. This is not the case in the former, where there is no steered dolly. While the effectiveness of this cost function is not as significant as the previous tractor-following cost function, it still managed to reduce the maximum swept path width to 3.99 meters.

A notable advantage of the axle mid-point cost function can be seen in the actuator performance. By comparing the plots in Figure 4.2a and 4.3a, we observe that the input dynamics under the axle mid-point cost function are significantly more refined, with fewer fluctuations and smoother transitions. This improvement suggests a more controlled and efficient steering response.




**(a)** Steering control input

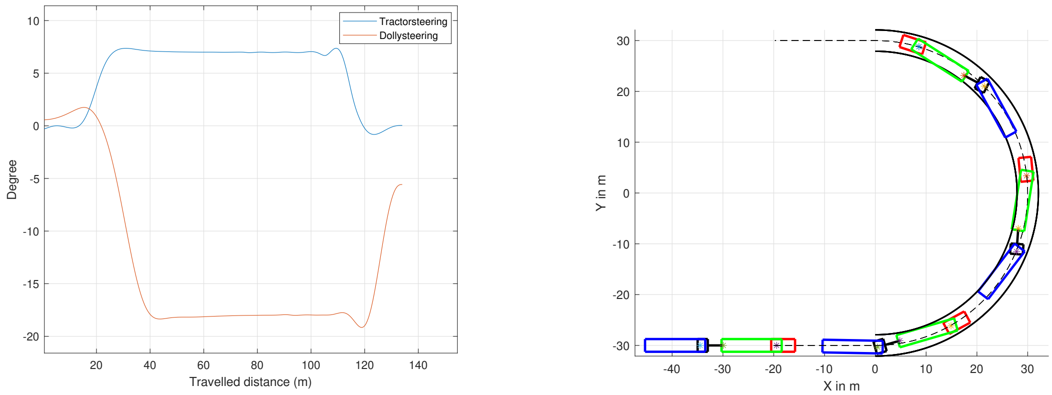
**(b)** Vehicle trajectory

**Figure 4.3:** Optimal control inputs and trajectory of vehicle using cost function 2

#### 4.1.1.5 Dolly Steering - Body Corner Cost Function

The performance of the body corner cost function in reducing swept path width closely mirrors that of the tractor-following cost function, achieving a reduced swept path of 3.84 meters, as previously noted. From the trajectory plots in Figures 4.2b, 4.3b, and 4.4b, it's evident that the articulation angles of the LCV are significantly more aggressive with the axle mid-point cost function compared to the other two approaches, particularly as the vehicle approaches the midpoint of the corner. This is clearly visible in the third instance of the LCV from the trajectory plot. While this aggressive articulation effectively minimizes the deviation of the first trailer, it has minimal impact on further reducing the overall swept path width.

Investigating the actuator dynamics, we observe that the gradient of the steering inputs in Figure 4.4a is quite similar to that in Figure 4.2a. However, one notable improvement in the body corner cost function is the elimination of fluctuations that were present in the tractor-following cost function. This comparison is made under similar maneuver conditions, with comparable weightages assigned to state deviations and input penalties across the different cost functions.




**(a)** Steering control input

**(b)** Vehicle trajectory

**Figure 4.4:** Control inputs and trajectory for vehicle using cost function 3

#### 4.1.1.6 Summary of Kinematic Results

By following the above steps, the evaluation of the cost functions for low speeds using the kinematic model was carried out for different maneuvers and the summary of their performance in reducing the swept path is presented in Figures from 4.5 to 4.7.

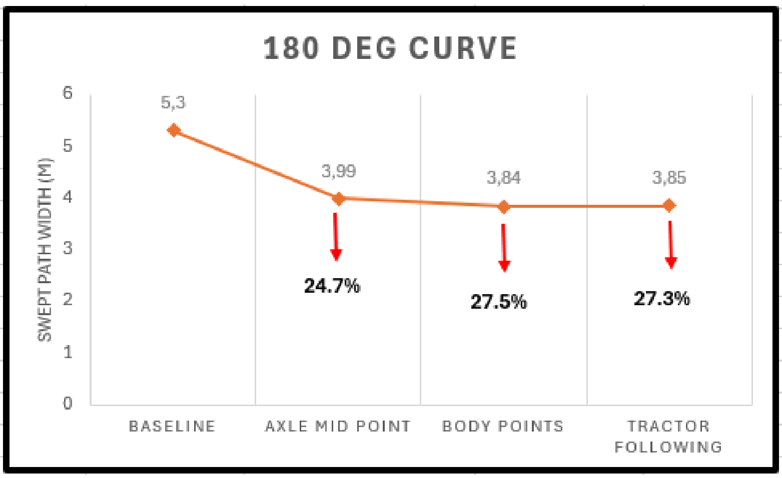



**Figure 4.5:** Performance of cost functions for a 180 deg. turn of radius 30m.

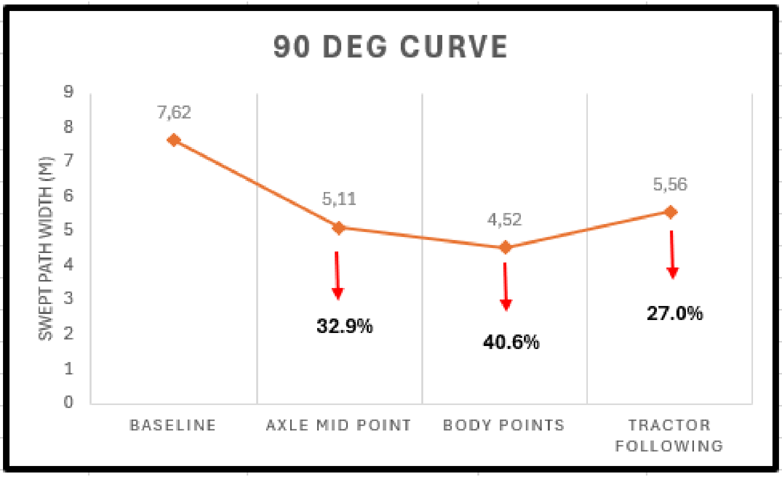



**Figure 4.6:** Performance of cost functions for a 90 deg. turn of radius 12.5m.

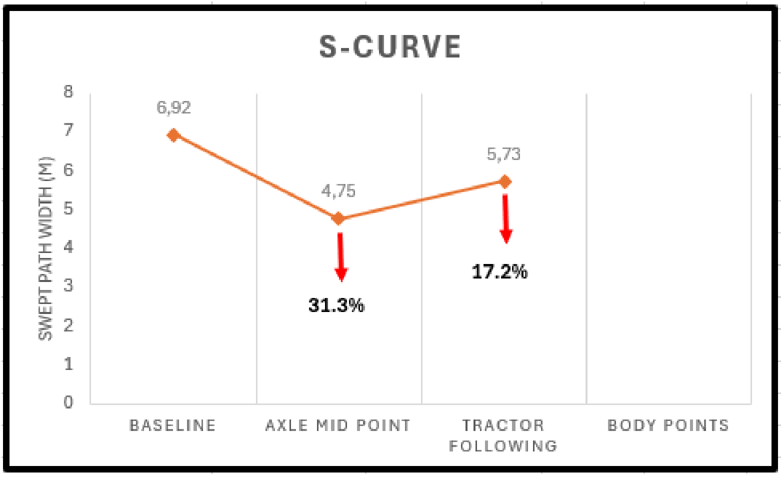
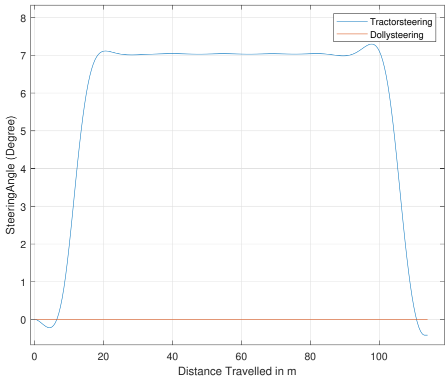



**Figure 4.7:** Performance of cost functions for a S-Shape maneuver of radius 20m.

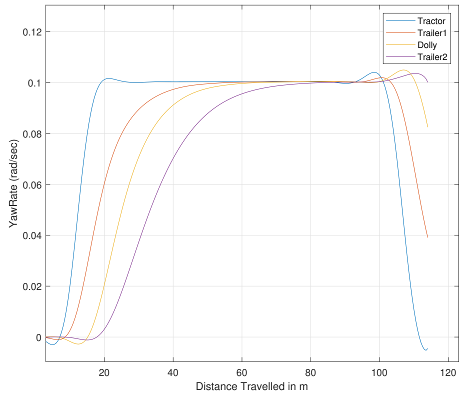
### 4.1.2 Kinetic Results

#### 4.1.2.1 No Dolly Steering

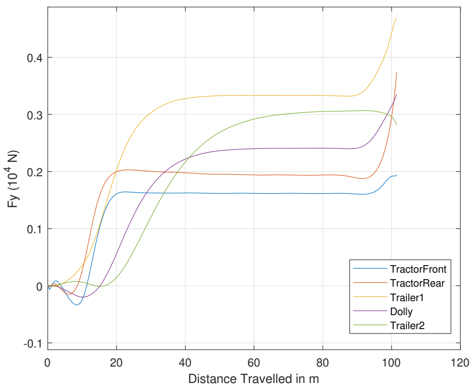
Figure 4.8 below illustrate the behavior of a vehicle when the steering control is not optimized, including the effects of dolly steering. Using the kinetic model, we analyzed the forces and yaw rates acting on each unit. A common observation is that the end of the maneuver is not smooth due to relaxed constraints, which simplifies the solution process. As shown in Figure 4.8c, the lateral force on units 2 and 4 are notably high because they carry heavy loads. Since the maneuver is performed at low speed, the yaw rates of each unit aim to achieve steady state due to constant steering. It's important to note that without dolly steering, the last unit is responsible for creating a high swept path width of 5.09m, which affects the vehicle's overall maneuverability. This insight highlights the significance of optimizing steering controls to improve vehicle performance.



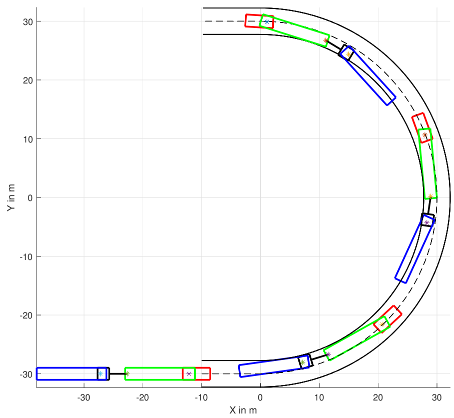

**(a)** Steering angle inputs




**(b)** Yaw rate outputs




**(c)** Lateral force acting on each units

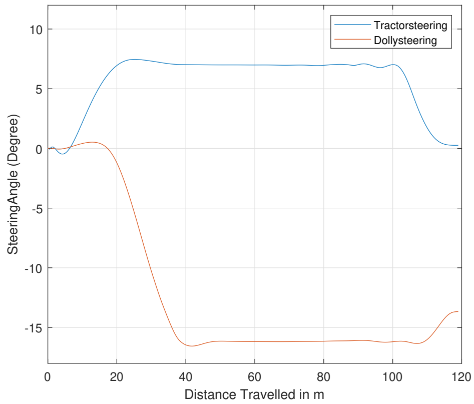



**(d)** Vehicle trajectory

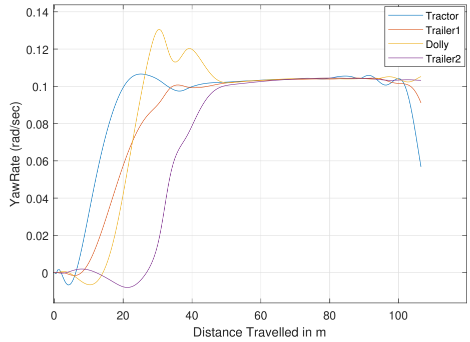
**Figure 4.8:** No dolly steering for 180 degree maneuver

#### 4.1.2.2 Dolly Steering - Axle Mid-point

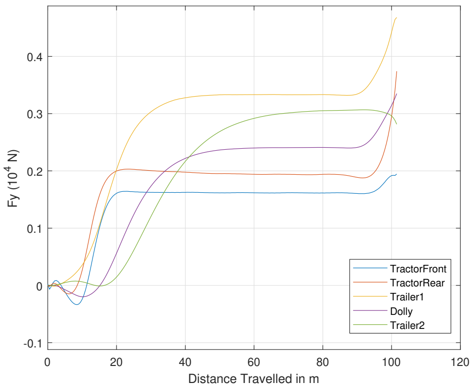
The plots below illustrate the vehicle's behavior with optimized steering allocations for both the tractor and dolly. With the dolly steering, there is a noticeable non-zero steering input in Figure 4.9a, leading to higher yaw rates and lateral forces, as shown in Figures 4.9b and 4.9c respectively. These forces stabilize as the steering input stabilizes, ultimately reducing the vehicle's swept path to 3.87m. Among the three cost functions tested, the axle midpoint cost function provided the best results for safety and maneuverability. Additionally, this optimized setup enhances overall vehicle control and efficiency during maneuvers.



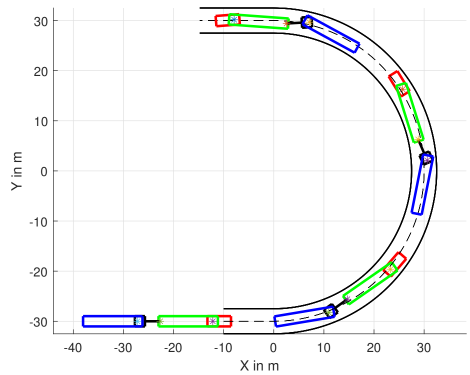

**(a)** Steering angle inputs




**(b)** Yaw rate outputs




**(c)** Lateral force acting on each units




**(d)** Vehicle trajectory

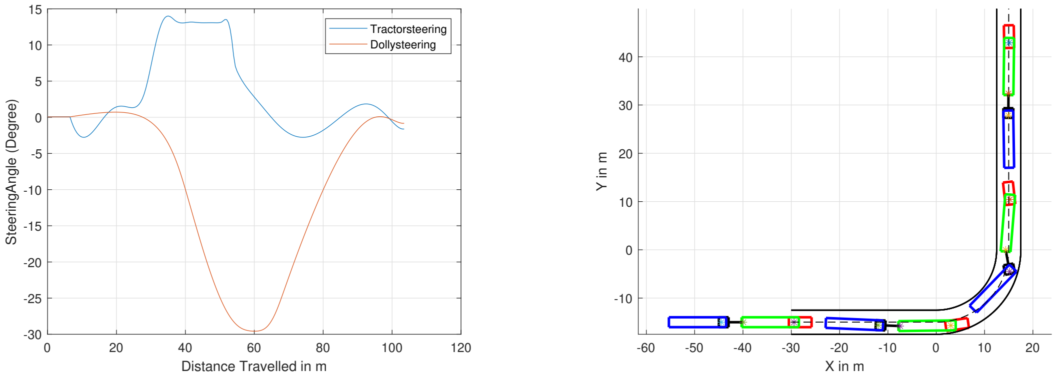
**Figure 4.9:** Optimized dolly steering for 180 degree maneuver

The results above indicate that the overall performance of the A-double combination using a steerable dolly, particularly for a 180-degree maneuver, is consistent regardless of the vehicle model used. Specifically, the swept path width and the steering angle values that produced this width are comparable across different models. However, to avoid making premature conclusions, we also investigated the performance for other maneuvers to ensure comprehensive analysis. This approach ensures a thorough understanding of the vehicle's behavior across various scenarios.

## 4.2 Comparison between Kinematic Model and Kinetic Model

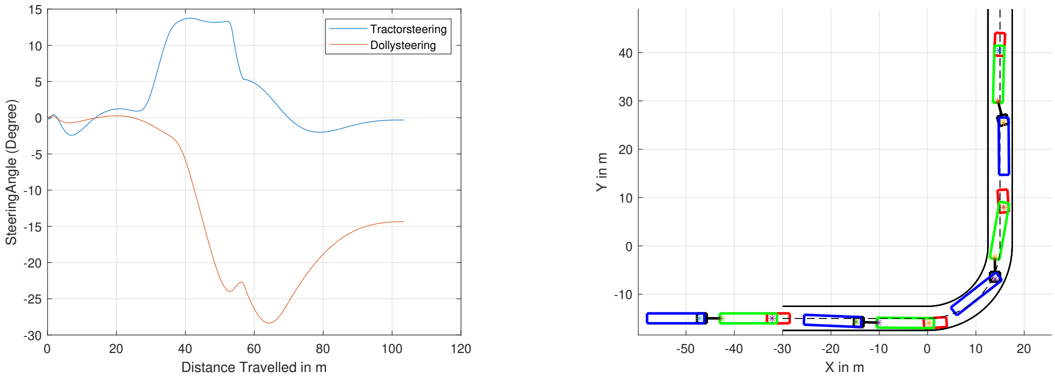
In this section, we will investigate and compare the results of optimizing steering input using two different vehicle models (kinematic and kinetic) for various maneuvers at low speeds. To ensure a fair comparison, we will primarily focus on the steering angles of the two units and the swept path created by the vehicle. This detailed comparison will help us understand the strengths and limitations of each model in

optimizing vehicle control.
Starting with the 90-degree maneuver, Figures 4.10a and 4.10b show the steering inputs and vehicle positions when using the kinematic model. Similarly, Figures 4.11a and 4.11b depict these characteristics when using the kinetic model. Although the steering input for the kinetic model is not smooth due to the high computational demands and time-consuming nature of solving the optimization problem, the results are still comparable. The swept path widths achieved after using different cost functions are compared for both kinematic and kinetic model in the table of comparison provided in Figure 4.14. Notably, during tighter turns, the vehicle tends to move outside the path before making the turn, a behavior observed with both models. This demonstrates the solver's effectiveness in finding optimal solutions for the given problem.




**(a)** Steering angle inputs **(b)** Vehicle trajectory

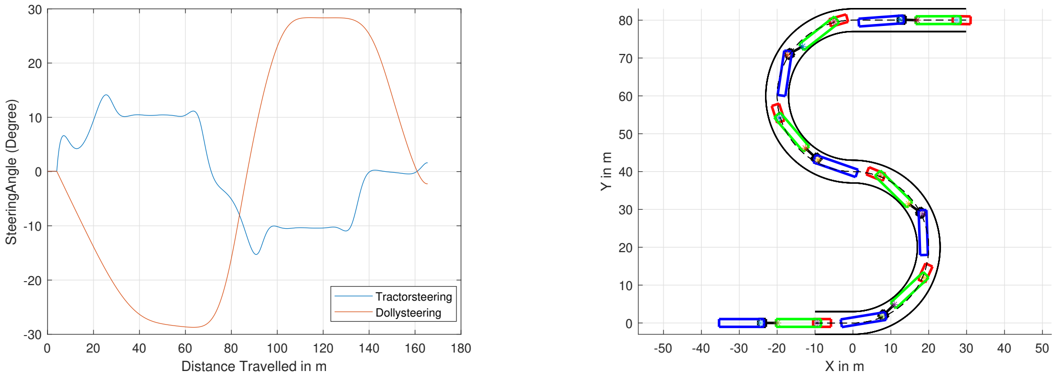
**Figure 4.10:** Optimal steering allocation using kinematic model for 90 degree turn




**(a)** Steering angle inputs **(b)** Vehicle trajectory

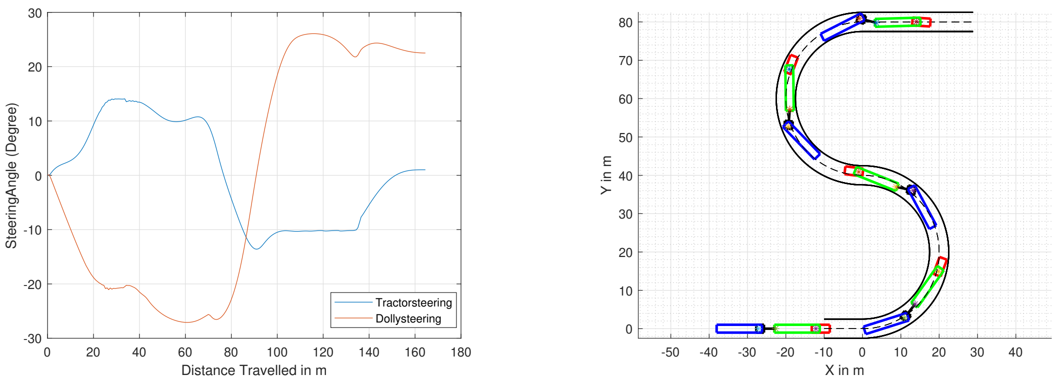
**Figure 4.11:** Optimal steering allocation using kinetic model for 90 degree turn

Moving to the S-curve maneuver, the comparison is similar to the 90-degree curve. Once again, the kinetic model did not produce as smooth results, but the outcomes from both models are similar, refer to Figures 4.12 and 4.13. For this maneuver, which required the most instances to optimize, the unit distance for each instance was larger and fixed to save time. Additionally, strict constraints were removed. Despite these adjustments, the trajectories obtained from both models are almost identical, demonstrating their effectiveness in handling complex maneuvers.




**(a)** Steering Angle Inputs **(b)** Vehicle Trajectory

**Figure 4.12:** Optimal steering allocation using kinematic Model for S curve




**(a)** Steering Angle Inputs **(b)** Vehicle Trajectory

**Figure 4.13:** Optimal steering allocation using kinetic Model for S curve

The comparison Table 4.14 summarizes all the results obtained using different cost functions when two different vehicle models were employed across various maneuvers. Although, some data is missing, as the solver either failed to find a solution or exceeded the anticipated number of iterations during optimization.

| MANOEUVRE | COST FUNCTION | SWEPT PATH WIDTH (m) | |
|---|---|---|---|
| | | Kinetic Model | Kinematic Model |
| 180 deg Curve | No Dolly Steering | 5.1 | 5.1 |
| | Axle Mid Point | 3.9 | 3.8 |
| | Body Corner Point | 3.8 | 3.7 |
| | Tractor Following | 3.8 | 3.9 |
| 90 deg. Curve | No Dolly Steering | 7.4 | 7.4 |
| | Axle Mid Point | 5.1 | 5.3 |
| | Body Corner Point | 4.5 | 4.6 |
| | Tractor Following | 5.5 | 5.9 |
| S Curve | No Dolly Steering | 6.7 | 6.7 |
| | Axle Mid Point | 4.7 | 4.5 |
| | Body Corner Point | 5.7 | 5.3 |
| | Tractor Following | 5.02 | - |

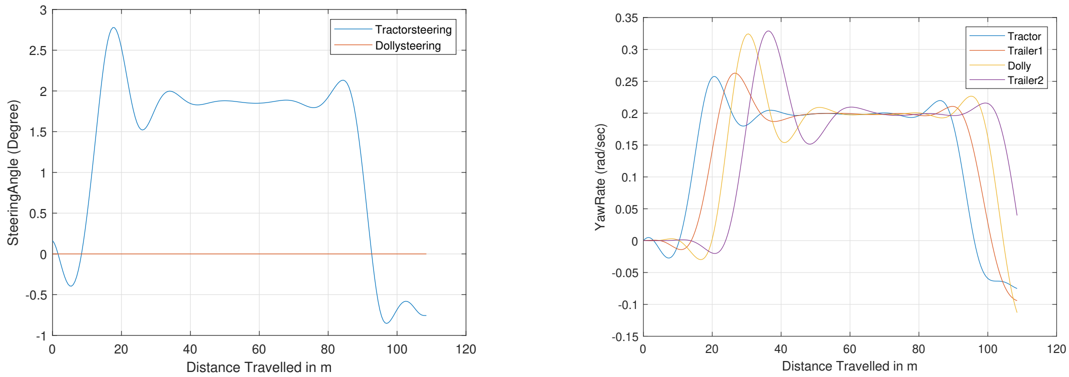
**Figure 4.14:** Comparison of Swept path values for Kinematic and Kinetic models at low speeds

## 4.3 Objective 2 Reflection - Minimizing Off-Tracking

Minimizing off-tracking at high speeds involves complex interactions between various dynamic forces acting on the vehicle. By utilizing a kinetic model and evaluating good cost function, it is possible to optimize the objective that significantly improve vehicle stability, handling, and safety during high-speed maneuvers.

### 4.3.1 High Speed Off-Tracking

Below in LCV, where only the tractor's front axle is steered and the dolly is not, the yaw rate exhibits distinct phases as the vehicle negotiates a curve, evident in Figure 4.15b . As the combination enters the curve, the steering angle of the tractor's front axle increases sharply, leading to a sudden spike in yaw rate. This spike is more pronounced in the rear units of the LCV due to their distance from the steering axle and hence creating higher lateral deviation on dolly and second trailer with respect to tractor's axle midpoint, as evident in Figure 4.16. During the long turn, the yaw rate stabilizes as the steering angle holds steady. Upon exiting the curve, the steering angle decreases, causing the yaw rate to drop rapidly as the vehicle realigns with the straight path. This dynamic behavior highlights the challenges in maintaining stability and control in LCVs during high-speed maneuvers.
To optimize the performance in terms of stability and maneuverability, dolly steering was introduced alongside tractor steering. Two cost functions, axle mid-point and tractor following, were evaluated. When the dolly was steered in combination with




**(a)** Steering angle inputs **(b)** Yaw rate outputs

**Figure 4.15:** High speed off-tracking without optimization

the tractor, along with the off-tracking, the yaw rate amplification was significantly reduced, leading to a smoother region of steady-state yaw rate, as shown in Figure 4.17b. This indicates that the objective of minimizing lateral off-tracking also minimizes yaw amplification, as the dynamics of the vehicle show interrelated effects on these objectives. Steering angles of both steerable units, as shown in Figure 4.17a, were in the same direction, effectively counteracting the centrifugal force and dragging the last unit inward. This coordinated steering reduced lateral off-tracking, as illustrated in the trajectory of the combination's improved performance, as shown in Figure 4.17.

Figure 4.18 below illustrates the overall reduction in off-tracking for each unit relative to the tractor, compared to using two distinct cost functions. Notably, the cost function based on the axle midpoint resulted in a greater reduction in off-tracking compared to the tractor-following cost function. This is because the former provided the tractor with slightly more steering freedom, allowing for optimized steering for both the tractor and dolly. In contrast, the latter cost function focused solely on optimizing the dolly's steering.

## 4.4 Objective 3 Reflection - Minimizing Rearward Amplification

To achieve the objective of reward amplification using different cost functions, no maneuver is more effective than a lane change. This maneuver effectively tests the vehicle's stability, handling, and responsiveness under dynamic conditions. The lane change requires precise control and coordination of steering inputs, making it an ideal scenario for evaluating and optimizing the performance of an LCV with both tractor and dolly steering.

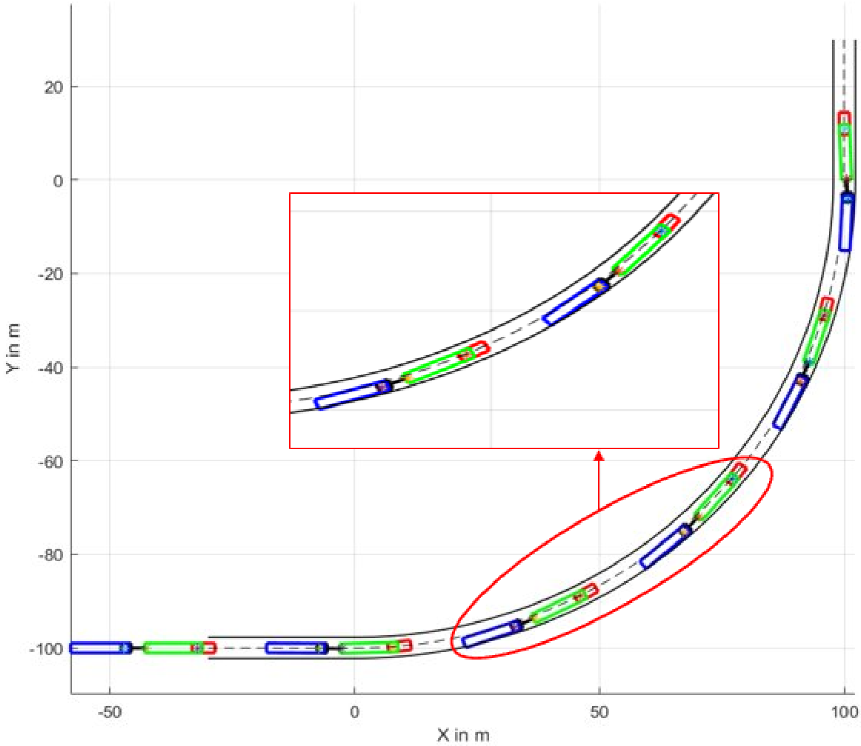


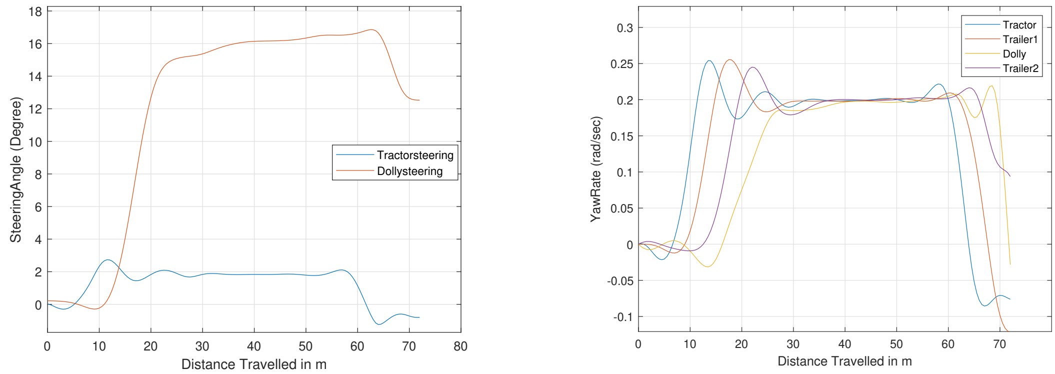

**Figure 4.16:** Vehicle trajectory without dolly steering - higher off-tracking

### 4.4.1 Lane Change

#### 4.4.1.1 Single Lane Change

Starting with a single lane change maneuver, the vehicle's performance is evaluated based on its response to a sudden steering input while traveling at 80 km/h. This maneuver is defined by the vehicle's ability to follow a desired path through a controlled steering input at the tractor's front axle. Figure 4.20b of the yaw rate plot illustrates the dynamics of this maneuver, showing that the last two units of the combination experience a significantly higher yaw rate compared to the tractor. The amplification is even more pronounced when all units are fully transitioned into the second lane, highlighting the challenge of maintaining stability and control during such maneuvers. This indicates that the yaw rate has amplified, with an amplification factor greater than 1, that is 1.55 and 1.69 for dolly and second trailer respectively.

When the dolly is steered in conjunction with the tractor, there is a significant reduction in yaw rate amplification. This factor was reduced by 36% and 31.3% for dolly and second trailer respectively. Such coordinated steering ensures better stability and control of the vehicle combination. The dolly's steering angle is typically in the opposite direction to that of the tractor and is higher in magnitude. This is to counterbalance the yaw forces acting on the last unit, thereby maintaining its yaw rate at a level similar to that of the tractor. As a result, the yaw rate of the entire vehicle combination in Figure 4.21b is more uniform, which reduces the overall yaw rate amplification.




**(a)** Steering angle inputs

**(b)** Yaw rate outputs

**Figure 4.17:** High speed off-tracking with optimization

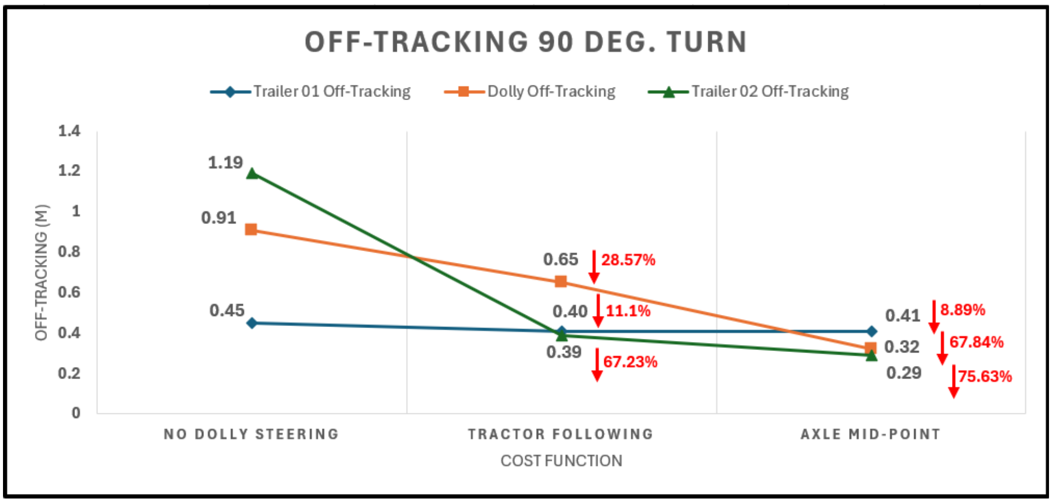



**Figure 4.18:** Performance of cost functions for reduction in off-tracking.

Figure 4.22 summarizes the reduction in the amplification factor for each unit. It is evident that, since the steering of the tractor was fixed in both steered and non-steered dolly vehicle combinations, there is no change in amplification for the first trailer. However, if the dolly were also propelled, it might have some effect on this factor.

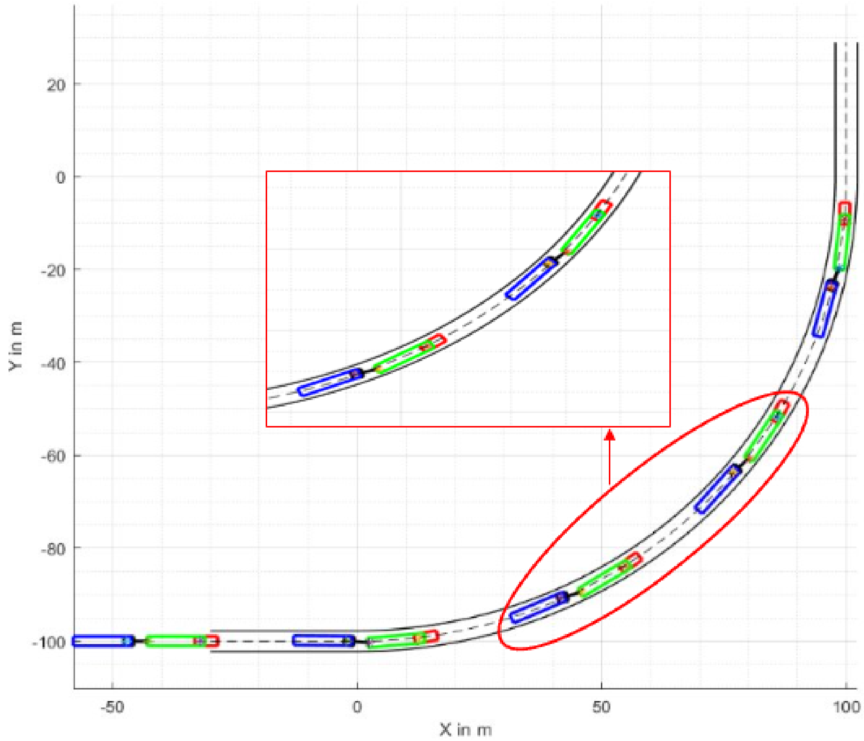



**Figure 4.19:** Vehicle trajectory after optimisation - reduced off-tracking

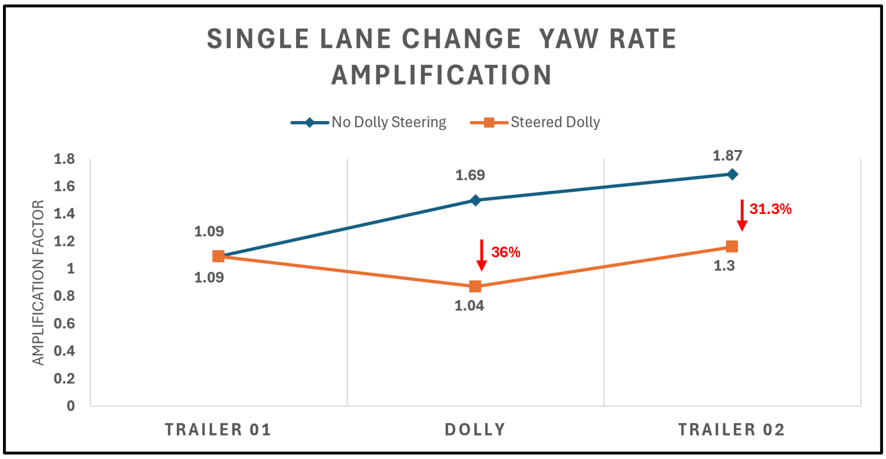
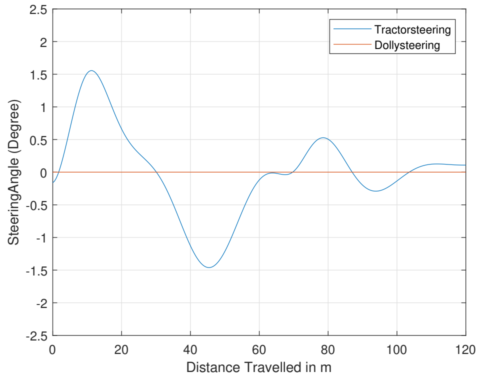


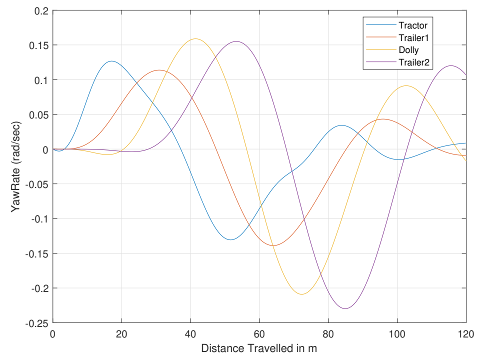

**Figure 4.22:** Reduction in rearward amplification for single lane change with dolly steering

#### 4.4.1.2 Double Lane Change

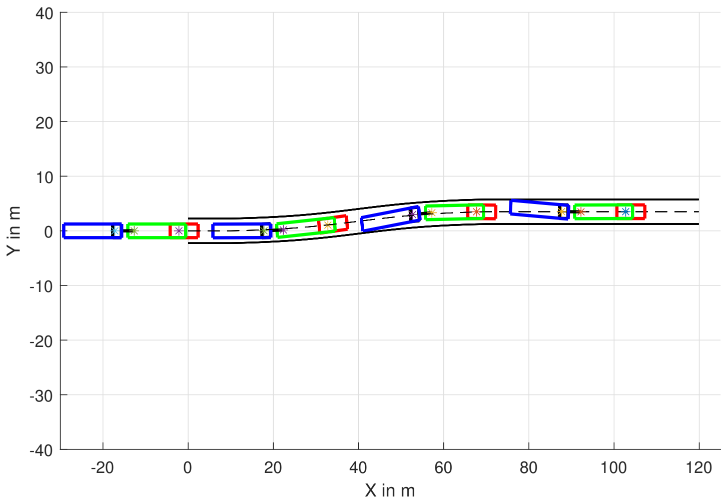
To further test the maneuverability and stability of the vehicle combination using optimized control allocation for yaw rate amplification reduction, a double lane change maneuver is performed.
The plot in Figure 4.23a below illustrates the front axle tractor steering angle, which exhibits a sinusoidal pattern due to the vehicle needing to steer twice. This complex maneuver imposes additional challenges on vehicle dynamics, as it requires rapid and




**(a)** Steering angle inputs



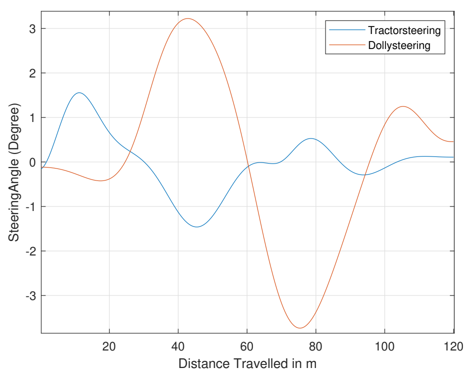

**(b)** Yaw rate outputs



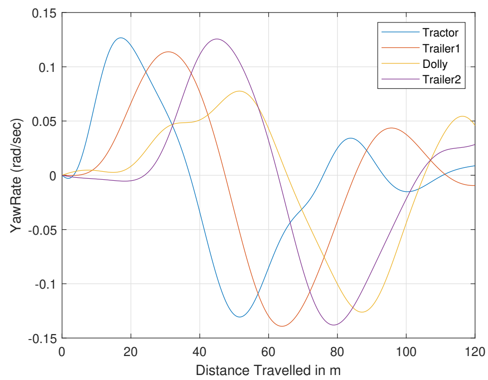

**(c)** Vehicle trajectory

**Figure 4.20:** Rearward amplification in single lane change when dolly is not steered

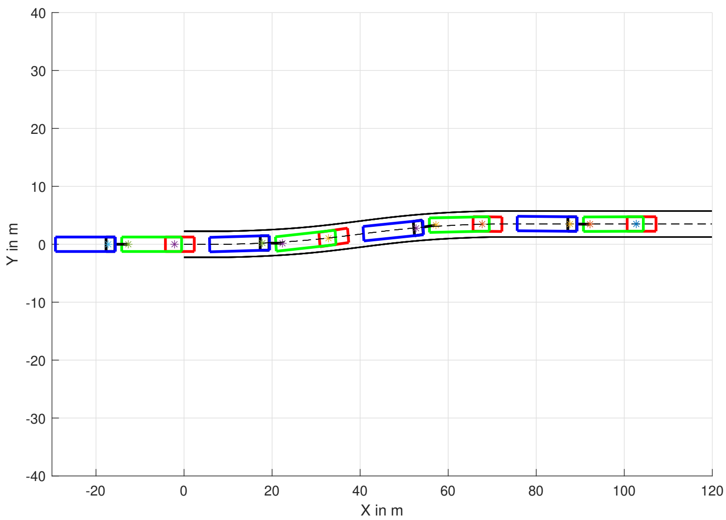
precise steering adjustments to maintain the desired path. As a result, the yaw rate experienced by the last two units of the combination is significantly higher, especially when the vehicle exits the curve. The yaw rate amplification factor for dolly and second trailer was observed to be 1.69 and 1.85 respectively. The rapid steering inputs and the necessity to change lanes twice amplify the yaw rate, indicating that the vehicle's stability is heavily tested under these conditions.
The amplified yaw rate highlights the importance of optimized control strategies to ensure that the last units of the combination can follow the tractor with minimal yaw rate discrepancies, thereby enhancing overall stability and maneuverability. Hence, when dolly steering is added to the vehicle combination, there is a significant reduction in yaw amplification for the dolly and trailer 2, by 38.4% and 30.4%, respectively, as shown in the figure. The synchronized steering between the tractor and dolly ensures that the yaw forces are better managed throughout the entire combination.




**(a)** Steering angle inputs




**(b)** Yaw rate outputs



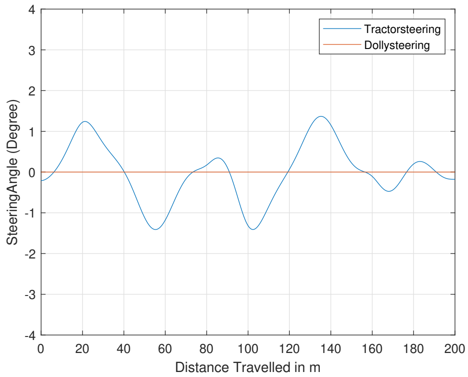

**(c)** Vehicle trajectory

**Figure 4.21:** Rearward amplification in single lane change when dolly is steered optimally

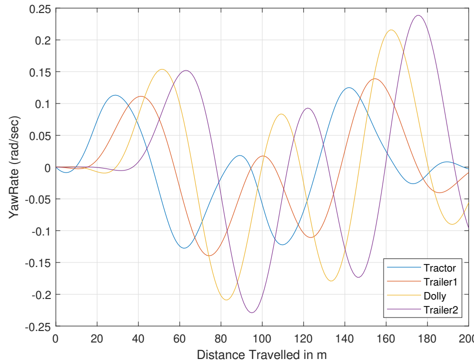
### 4.4.2 Dolly with steerable and propelled axle

In previous sections, we discussed how the integration of a steerable dolly axle in an LCV effectively reduces swept path, offtracking, and rearward amplification. We then explored the potential benefits of enhancing this configuration by adding propulsion to the dolly axle, making it not only steerable but also capable of self-propulsion.

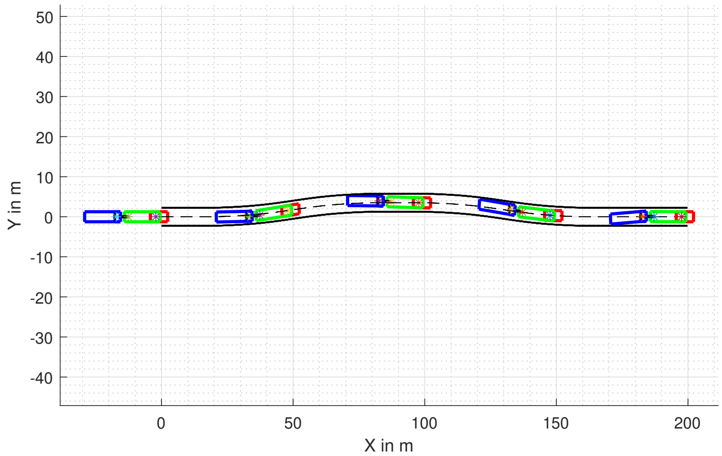
The addition of propulsion to the steerable dolly axle is expected to improve control over both the dolly and the trailing second trailer, especially during high-speed maneuvers. To test this hypothesis, we conducted an optimization analysis focusing on a high-speed 90-degree turn, using a dolly axle equipped with both steering and propulsion capabilities. The primary objective was to assess the impact of this




**(a)** Steering angle inputs




**(b)** Yaw rate outputs




**(c)** Vehicle trajectory

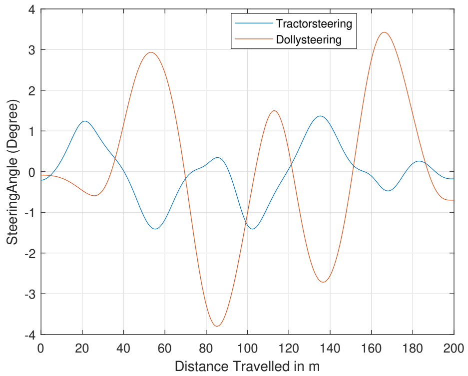
**Figure 4.23:** Rearward amplification in double lane change when dolly is not steered

configuration on reducing offtracking and yaw amplification.

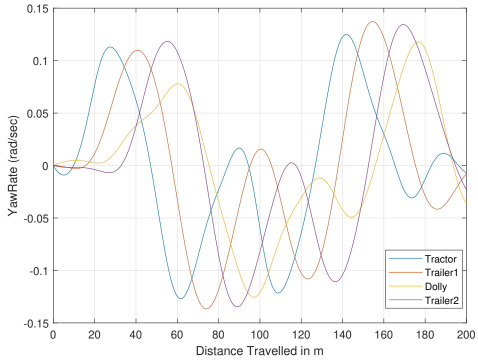
The comparative results, illustrating the reduction in high-speed offtracking when using a steerable dolly versus a dolly with both steering and propulsion, are presented in Figure 4.27.

From the plot in Figure 4.27, it's evident that while a dolly equipped with both steering and propulsion does reduce off-tracking compared to a non-steered dolly, the improvement does not surpass the benefits observed with a dolly that has a steerable axle alone. The potential for enhanced maneuverability through the use of a propelled axle would likely be greater if differential braking capabilities were available, allowing independent control of the propulsion forces on the left and right wheels. However, this optimization falls outside the scope of our current single-track model.

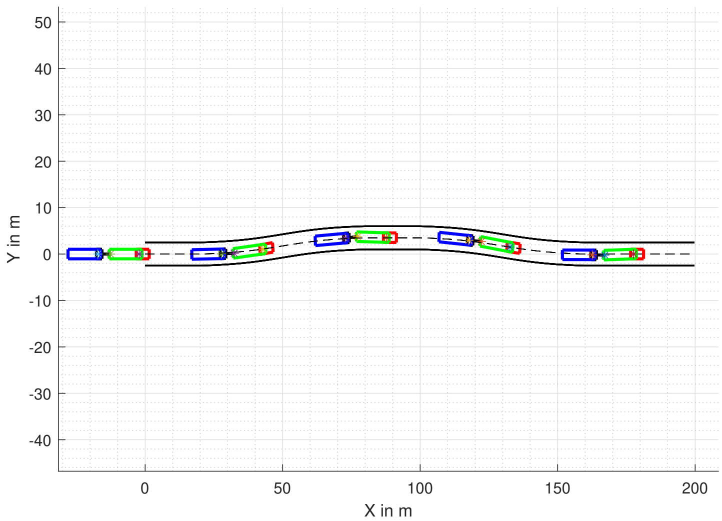
When examining the trajectory of the LCV shown in Figure 4.28, and comparing it with the trajectory of the LCV equipped with only a steerable axle (as seen in Figure 4.19), it is evident that the LCV with the additional capabilities adheres more effectively to the road boundaries. Specifically, it maintains a greater distance




**(a)** Steering angle inputs




**(b)** Yaw rate outputs




**(c)** Vehicle trajectory

**Figure 4.24:** Rearward amplification in double lane change when dolly is steered optimally

from the road edges, demonstrating a more substantial compliance with the road limits at higher speeds.